REIDEMEISTER-II COLLAPSE OF TWO-CHANGE LOCAL PLAT FAMILIES FOR UNKNOTTING AND CONWAY MUTATION

ARGUS OPEN RESEARCH PROJECT

Abstract. Kirby Problem 1.3 asks how unknotting number behaves under connected sum and Conway mutation. We prove a structural exclusion theorem for the local plat tangle-mutation family: for both marked crossing changes and under both horizontal and vertical Conway half-twists, the mutated plat closure reduces by an immediate Reidemeister-II move to an unknotting-number-one closure, ruling out a gap witness in that family.

1. Main Reduction

Proposition 1.1. *In the four-strand local plat family with two marked crossing changes inside the Conway sphere, performing either half-twist involution pairs the modified crossing with an adjacent oppositely signed plat bridge, allowing a single Reidemeister-II cancellation to unknot the mutated closure in one crossing change.*

Date: August 29, 2026.
2020 *Mathematics Subject Classification.* Primary 57K10; Secondary 57M25.